RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

www.rvce.edu.in
Tel: +91-80-68188100
+91-80-68188111
+91-80-68188112

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERIG

WildSentinel: Safeguarding the Human-Wildlife Interface

Experiential Learning

REPORT

Submitted by

Amogh S 1RV24IS013

Arryan Lodha 1RV24IM064

Avyav Saraf 1RV24IS020

Under the guidance of

Dr. M Rajeswari

Assistant Professor, Department of Biotechnology

In partial fulfilment for the award of degree of

Bachelor of Engineering

For the academic year

2025 – 2026

1. Introduction

Zoonotic diseases, which spread from animals to humans, have become a major global health concern. Recent outbreaks such as COVID-19, Ebola, SARS, and Nipah have demonstrated how rapidly infectious diseases can spread across countries. Human activities such as deforestation, habitat destruction, wildlife trafficking, urban expansion, and environmental degradation have significantly increased interactions between humans and wildlife, creating favorable conditions for disease spillover.

Traditional surveillance systems are largely reactive, identifying outbreaks only after they occur. Such systems often focus on individual factors like climate, biodiversity, or wildlife trade separately, limiting their ability to predict emerging threats. There is therefore a need for an integrated predictive system capable of combining ecological, environmental, and socio-economic factors to identify high-risk regions before outbreaks occur.

WildSentinel addresses this challenge by integrating wildlife trade records, species distribution data, satellite-derived environmental indicators, historical outbreak information, and biosafety metrics into a unified predictive framework. Using machine learning, geospatial analytics, and network-based modeling, the system predicts zoonotic spillover risks and provides actionable insights for policymakers, researchers, and public health authorities.

---

2. Problem Definition

2.1 Problem Statement

Emerging zoonotic diseases pose a significant threat to global public health. Increasing habitat fragmentation, wildlife trade, and environmental degradation have intensified human-wildlife interactions, raising the likelihood of pathogen spillover events. Existing monitoring systems lack the ability to integrate multiple risk factors and provide early warnings. The challenge is to develop a predictive framework capable of identifying geographical regions at high risk of future zoonotic outbreaks.

2.2 Background Information (Literature Review)

Several studies have established strong relationships between environmental disturbances and zoonotic disease emergence. Research indicates that deforestation and wildlife trafficking increase opportunities for pathogen transmission between animals and humans. Habitat fragmentation and biodiversity loss further elevate spillover risks by increasing species interactions.

International organizations such as the OECD and UNEP emphasize the importance of preventive and One Health approaches to address future pandemics. Advances in machine learning, geospatial technologies, and biosurveillance have enabled the development of predictive systems capable of analyzing large ecological datasets. However, most existing approaches focus on isolated factors rather than integrating wildlife trade, biodiversity, environmental degradation, and outbreak histories into a single predictive framework.

WildSentinel bridges this gap by combining multiple data sources and advanced analytical techniques to provide comprehensive zoonotic risk assessment.

---

3. Objectives

3.1 Primary Objectives

1. Predict zoonotic spillover risks using ecological and anthropogenic data.
2. Identify geographical hotspots vulnerable to future outbreaks.
3. Develop an interactive geospatial risk visualization platform.
4. Support evidence-based biosafety and conservation decisions.
5. Provide an early warning system for disease emergence.

3.2 Secondary Objectives

1. Analyze the impact of wildlife trade on disease emergence.
2. Evaluate environmental factors contributing to spillover risk.
3. Simulate intervention strategies using Monte Carlo methods.

4. Promote the One Health approach for sustainable coexistence between humans and wildlife.

---

4. Methodology

4.1 Approach

WildSentinel follows a data-driven predictive modeling approach integrating environmental, ecological, and socio-economic datasets.

Steps Involved:

1. Collection of wildlife trade, biodiversity, environmental, and outbreak datasets.
2. Data cleaning, normalization, and spatial alignment.
3. Extraction of ecological and anthropogenic features.
4. Construction of network-based analytical relationships.
5. Training and validation of XGBoost predictive model.
6. Scenario simulation using Monte Carlo methods.
7. Visualization of risk hotspots through interactive maps.

4.2 Procedures

Phase 1: Data Collection

- Collection of CITES wildlife trade records.
- Acquisition of GBIF species occurrence datasets.
- Retrieval of NASA MODIS satellite imagery.
- Collection of WHO-WAHIS outbreak records.

Phase 2: Data Processing

- Data cleaning and standardization.
- Geospatial alignment.

- Raster image processing.
- Temporal aggregation.

Phase 3: Model Development

- Feature engineering.
- Network graph construction.
- XGBoost model training.

Phase 4: Validation and Analysis

- Testing using unseen data (2021–2024).
- Scenario simulations.
- Interactive dashboard generation.

---

5. Project Execution

5.1 Planning and Design

The project began with identifying major zoonotic risk drivers through literature review and expert consultation. Multiple datasets were evaluated based on accessibility, relevance, and reliability. The system architecture was designed to support large-scale geospatial processing and predictive analytics.

Design considerations included:

- Scalability
- Data integration capability
- Real-time visualization
- Decision-support functionality

5.2 Implementation

The implementation involved:

- Dataset acquisition and preprocessing.

- Development of feature extraction pipelines.
- Construction of predictive machine learning models.
- Deployment of geospatial visualizations.
- Integration of scenario simulation modules.

The final system generated risk prediction outputs and displayed them through an interactive dashboard.

---

6. Tools and Techniques Used

6.1 Tools

Tool	Purpose
CITES Database	Wildlife trade records
GBIF	Species distribution data
NASA MODIS	Satellite imagery and environmental indicators
WHO-WAHIS	Historical outbreak data
Python	Data processing and model development
XGBoost	Predictive modeling
Leaflet.js	Interactive mapping
GIS Tools	Spatial analysis

6.2 Techniques

Data Preprocessing

Cleaning, normalization, geocoding, and temporal aggregation.

Feature Engineering

Creation of Trade Intensity Index, Forest Fragmentation Index, Species Richness Index, and Environmental Stress Indicators.

Machine Learning

XGBoost classification for spillover risk prediction.

Monte Carlo Simulation

Risk assessment under different intervention scenarios.

Geospatial Analytics

Identification and visualization of outbreak hotspots.

---

7. Results and Discussion

7.1 Final Results

The developed model demonstrated strong predictive capability on unseen data.

Model Performance

Metric	Value
AUC-ROC	0.8742
AUC-PR	0.6704
True Positives	4176
False Negatives	751
False Positives	3903
True Negatives	11449

Feature Importance

Rank	Feature	Importance
1	Total Export Events	36.08%
2	Total Import Events	19.92%
3	Total Live Exports	9.83%
4	Total Live Imports	7.39%
5	Percent Forest	5.70%



Figure 1 Zoonotic Spillover Heatmap

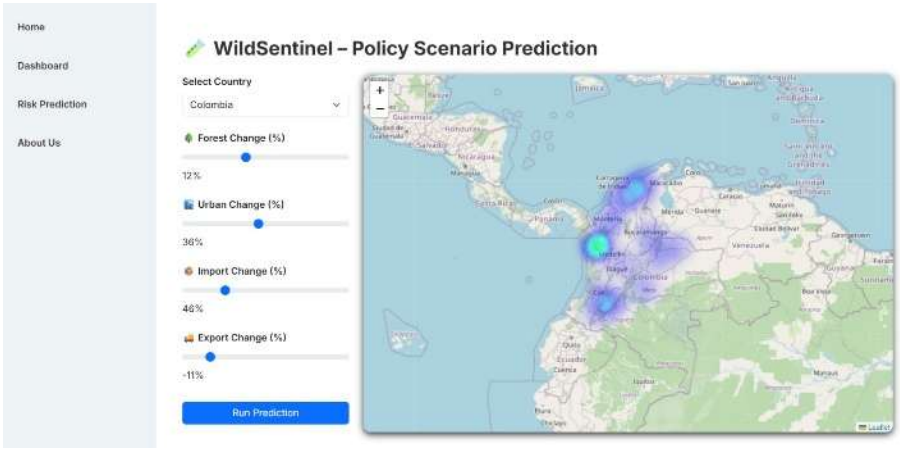


Figure 2 Predicting Spillover risk as per Given Scenerio

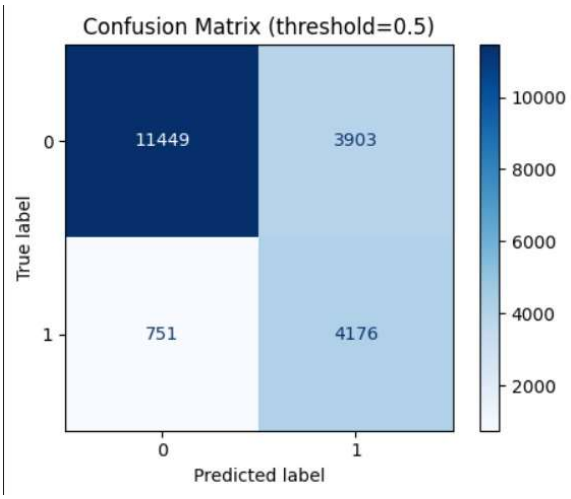


Figure 3 Confusion Matrix for the predicted True or False analysis

7.2 Discussion

The results reveal that wildlife trade and habitat degradation are the strongest predictors of zoonotic spillover risk. Regions experiencing intense trade activity combined with rapid environmental degradation exhibited significantly higher risk scores.

Monte Carlo simulations showed that reducing illegal wildlife trade and increasing forest restoration efforts could substantially reduce outbreak probability. The findings validate the effectiveness of integrated predictive modeling for bio-surveillance and public health preparedness.

---

8. Prototype (Software)

8.1 Prototype Description

WildSentinel is a web-based predictive bio-surveillance platform designed to identify zoonotic spillover hotspots.

Key Features

- Risk prediction engine
- Interactive heatmap visualization
- Scenario simulation dashboard

- Geospatial analytics
- Outbreak trend analysis

8.2 Development Process

The prototype was developed through iterative stages involving data integration, model development, validation, and dashboard deployment.

Challenges Faced

- Large-scale geospatial data processing.
- Inconsistent biodiversity records.
- Balancing highly imbalanced outbreak datasets.

Solutions

- Spatial data optimization.
- Extensive preprocessing.
- Pseudo-absence generation for model training.

8.3 Testing and Validation

The system was validated using unseen outbreak data from 2021–2024. The model achieved an AUC-ROC score of 0.8742, demonstrating strong predictive performance.

Cross-validation and scenario testing confirmed the reliability of risk predictions.

---

9. Conclusion

9.1 Summary

WildSentinel successfully integrates ecological, environmental, and wildlife trade datasets into a unified predictive framework for zoonotic spillover assessment. The developed system accurately identifies emerging hotspots and provides valuable decision-support capabilities for biosafety planning and disease prevention.

The project demonstrates how machine learning and geospatial analytics can strengthen global health preparedness through early warning systems and targeted interventions.

9.2 Personal Reflection

Working on WildSentinel provided valuable experience in applying machine learning, geospatial analytics, and biosurveillance concepts to solve a real-world public health challenge. The project improved our understanding of interdisciplinary problem solving, data integration, predictive modeling, and the importance of the One Health approach. It also enhanced our teamwork, research, and technical implementation skills.